SHORT · SPARK SQL + SAFE DELTA

Spark SQL and a safe managed Delta write

Spark SQL lets you query the same DataFrame with familiar SQL. Register a temp view, then SELECT, WHERE, and GROUP BY. Results match the DataFrame API.

Key points

- Actual SQL aggregation matched five region/category buckets and 142.50 total.
- Created unique schema workspace.bt_query_cf911148fc3d.
- Wrote managed Delta table region_category_revenue with mode("error").
- PASS: 5 buckets / 6 orders / revenue 142.50.

Tiny example

Validated Part 3 write. mode("error") fails instead of overwriting.

```
summary.write.format("delta") \
  .mode("error") \
  .saveAsTable(table_fqn)
# Actual PASS: 5 buckets / 6 orders / 142.50
# FQN: workspace.bt_query_cf911148fc3d.region_category_revenue
```

Misconception check

mode("error") is not the same as overwrite. If the table already exists, the write fails. That protects your first write from an accidental wipe.

Remember

SQL and DataFrames agree, and mode error keeps your first write safe.

Validated on Databricks Free Edition with serverless compute and Unity Catalog. Managed Delta tables and mode("error") work on Free Edition.